`scripts/autonomous-qa/run-matrix.sh` — User Guide

Last verified: 2026-07-26 (LVA-018 script-docs backfill cycle)
Inheritance: HelixConstitution §11.4.18 (script docs), §11.4.128 (evidence layout); Lava §6.AG / §6.AH / §6.X (containerized emulators), §6.J (anti-bluff verdicts) **Classification:** project-specific

Overview

Orchestrates the autonomous-QA backend × provider × query matrix — **ONE backend at a time** (operator rule: never run both APIs at once):

```
backend up → boot ONE containerized KVM emulator (kept alive)
  → for each provider-subset × query:
      FRESH install + record + run Challenge70 (via run-
iteration.sh)
  → tear emulator down → backend down → write curated summary.md
```

Plan: `docs/superpowers/plans/2026-06-29-autonomous-qa-backend-provider-matrix.md`.

Usage

```
scripts/autonomous-qa/run-matrix.sh --backend goapi|apiapp \
    [--queries 1080p,mp3] [--subsets all] [--external-backend]

# Keystone (Phase 0): single provider, single query
scripts/autonomous-qa/run-matrix.sh --backend goapi --subsets
        rutracker --queries 1080p
```

Flag	Meaning
`--backend goapi|apiapp`	Which API backend the on-device client targets (required)
`--queries <csv>`	Search queries to run per subset (default `1080p,mp3`)
`--subsets all|<csv1>;<csv2>`	`all` = every non-empty subset of the API-backed providers (31 subsets, see `lib-`

Flag	Meaning
	`subsets.sh`); otherwise `;`-separated CSV groups, e.g. `"rutracker;kinozal,nnmclub"`
`--external-backend`	Assume the backend is already up + healthy out-of-band (e.g. thinker via `distribute-api-remote.sh`); the script skips bring-up/teardown

Only **API-backed** provider ids (`QA_PROVIDERS` in `lib-subsets.sh`: rutracker, nnmclub, kinozal, archiveorg, gutenberg) are honored. `rutor` (bundled-only) and `iptorrents` (Jackett-only) are dropped with a warning — neither API backend vends them via `GET /v1/providers`, so Challenge70 would only SKIP opaquely inside the emulator.

Evidence layout

```
.lava-ci-evidence/autonomous-qa/<date>/<backend>/
  summary.md                  # curated matrix table + totals
  <subset-slug>-<query>/      # per-iteration evidence (see run-iteration.sh)
    verdict.json  junit.xml  raw/
```

Anti-bluff properties (§6.J)

- Any stale `verdict.json` is deleted **before** each iteration, so a prior-run PASS can never be misread when `run-iteration.sh` dies early (sixth-law-incidents 2026-07-03 missing-APK/stale-verdict PASS bluff).
- An iteration with no `verdict.json` is recorded **FAIL** (a setup error is never a pass — no test executed).
- A non-zero iteration exit code can **never** be counted PASS, even if the verdict file says PASS (defense-in-depth).
- Unknown/garbled verdict tokens classify as FAIL (`qa_classify`).

Exit codes

- `0` — matrix ran to completion (per-iteration verdicts live in `summary.md`; the script itself does not fail on iteration FAILs)
- `2` — usage error, unknown backend, or no API-backed subsets remain after filtering

Companion files

- `scripts/autonomous-qa/run-iteration.sh` — one matrix cell (install, record, run, verdict)

- `scripts/autonomous-qa/lib-subsets.sh` — provider-subset enumeration
- `scripts/autonomous-qa/lib-emulator.sh` — containerized KVM emulator lifecycle
- `scripts/autonomous-qa/lib-backend.sh` — backend bring-up / targeting / teardown
- `scripts/autonomous-qa/lib-summary.sh` — verdict parsing / classification helpers
- `scripts/autonomous-qa/aggregate-evidence.sh` — rolls the matrix up into the §6.AK distribute-gate artifacts